"If you know the enemy and know yourself, you need not fear the result of a hundred battles." — *Sun Tzu*

Race to 2026

Two players start with the number 0. On each turn, a player adds an integer from 1 to 10 (inclusive) to the current sum. The player who reaches exactly 2026 wins.

Question: Who has a winning strategy? Describe it explicitly.

Source: Based on Bachet's Game

Breaking the Bar

You have a rectangular bar of chocolate consisting of $m \times n$ small squares. Two players take turns breaking the bar (or any piece of it) along the lines separating the squares. The player who makes the last break, leaving only 1×1 squares, wins.

Question: Which player has a winning strategy, the first or the second? (The answer depends on the values m, n.)

Source: Folklore

Classic Nim

There are three piles of coins with sizes 3, 5, and 7. Two players take turns choosing one pile and removing any positive number of coins from it (they can take the whole pile). The player who takes the last coin wins.

Question: Does the first player have a winning strategy? If so, what is the first move?

Source: Classic

Game of Chomp

A chocolate bar is a rectangle of $m \times n$ squares. The top-left square $(1, 1)$ is poisoned. Two players take turns picking a remaining square (i, j) (other than the poisoned one) and eating it along with all squares below and to the right of it (i.e., all squares (r, c) with $r \geq i$ and $c \geq j$). The player forced to eat the poisoned square loses a life.[a]

Question: Do you want to play first, or second?

Source: David Gale

[a] Don't worry: It's a videogame!

Keep it nonsingular!

Alan and Barbara play a game in which they take turns filling entries of an initially empty 2026×2026 array. Alan plays first. At each turn, a player chooses a real number and places it in a vacant entry. The game ends when all the entries are filled. Alan wins if the determinant of the resulting matrix is nonzero; Barbara wins if it is zero.

Which player has a winning strategy?

Source: Adapted from Putnam 2008 A2

The "Short-of-Prime" Game

Alice and Bob play a game in which they take turns removing stones from a heap that initially has n stones. The number of stones removed at each turn must be one less than a prime number. The winner is the player who takes the last stone. Alice plays first. (For example, if n = 17, then Alice might take 6 leaving 11; then Bob might take 1 leaving 10; then Alice can take the remaining stones to win.)

Prove: There are infinitely many n such that Bob has a winning strategy.

Source: Putnam 2006 A2

Polynomial Coefficients

Consider the polynomial $P(x) = x^{2026} + a_{2025}x^{2025} + \cdots + a_1 x + a_0$. David and Goliath take turns choosing the coefficients $a_{2025}, \ldots, a_0$. David goes first. Each coefficient must be a real number. The game ends when all coefficients correspond to a fixed real number. David wins if the polynomial $P(x)$ has **no** real roots. Goliath wins if $P(x)$ has at least one real root.

Question: Who has a winning strategy?

Source: Putnam 1999 A3

Determinant Tic-Tac-Toe

In Determinant Tic-Tac-Toe, Player 1 enters a 1 in an empty 3×3 matrix. Player 0 counters with a 0 in a vacant position, and play continues in turn until the 3×3 matrix is completed with five 1's and four 0's. Player 0 wins if the determinant is 0 and player 1 wins otherwise.

Question: Who has a winning strategy? Describe it explicitly.

Source: Putnam 2004 A4

The Angel and the Devil

An Angel and a Devil play on an infinite chessboard. The Angel has wings and can jump to any square within distance K king's moves (for some large fixed integer K). The Devil, on his turn, can destroy any single square of the board (except the one the Angel is currently on). The Angel aims to keep moving forever; the Devil aims to strand the Angel by surrounding it with destroyed squares.

Question: Can the Devil always trap the Angel for sufficiently large K?
(This was a famous open problem solved only recently!) en.wikipedia.org/wiki/Angel_problem

Source: Conway / Berlekamp